Retail Pulse – AI-Powered Customer Analytics & Demand Forecasting

Transforming Retail Intelligence with AI-Driven Insights

Zidio Internship Project

Developed By: Team 20

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Project Duration:25th June-25th July

Project Overview

Retail Pulse is an AI-powered retail analytics platform designed to help businesses understand customer behaviour, forecast product demand, and optimize inventory management using Machine Learning and MLOps practices. The project focuses on improving business intelligence through predictive analytics and real-time visual dashboards.

Vision & Objectives:

• Predict future product demand using AI forecasting models.

• Analyse customer purchasing behaviour.

• Identify churn-prone customers.

• Improve inventory planning and reduce overstocking.

Target Users / Use Cases:

• Retail store managers

• E-commerce businesses

• Supply chain teams

• Business analysts and decision-makers

Business Value Delivered:

• Better forecasting accuracy

• Reduced inventory wastage

• Improved customer retention

• Data-driven decision making

Non-functional Goals:

• Scalable ML pipeline

• Low latency dashboard visualization

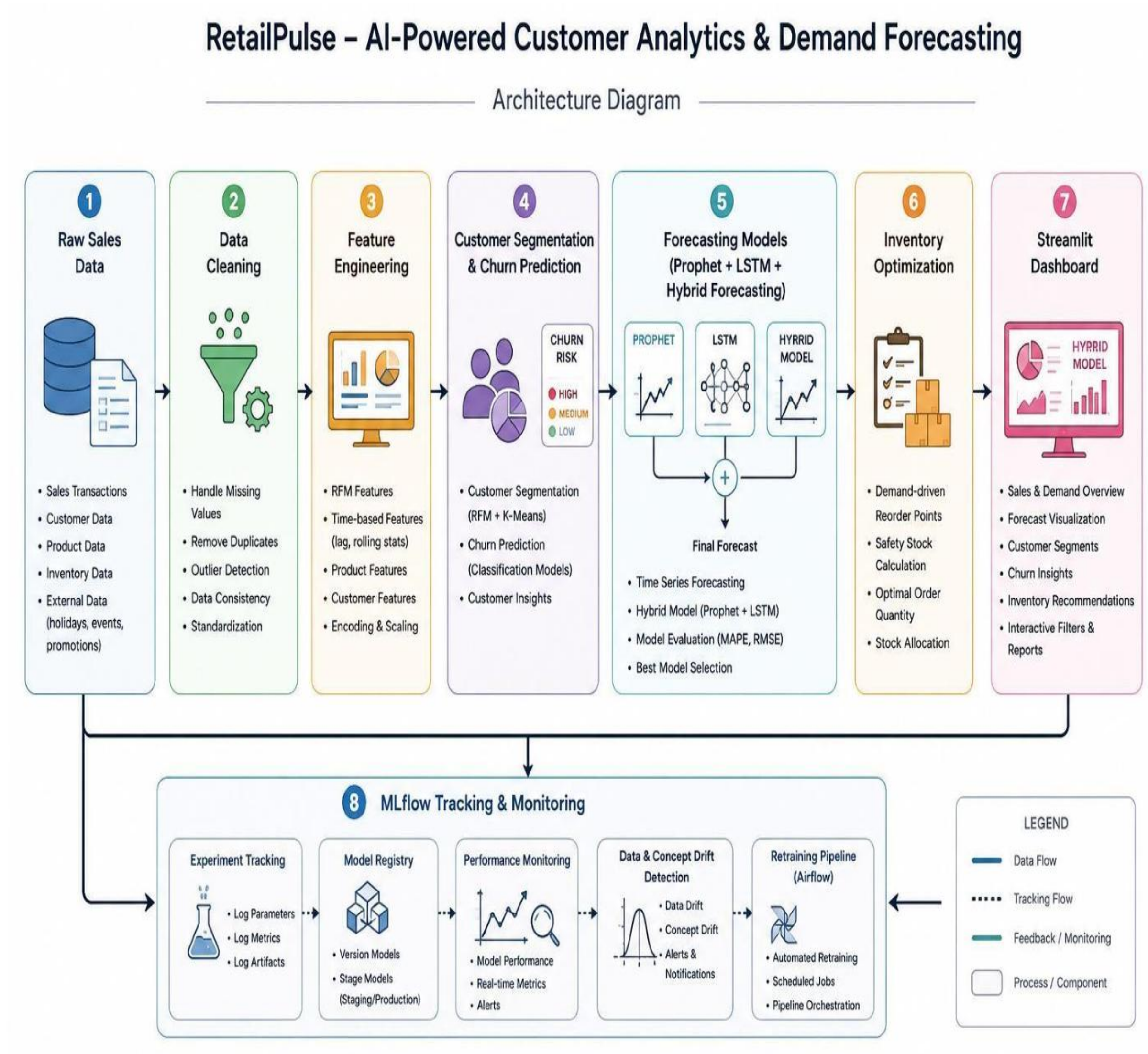
• Reliable model tracking and monitoring

• Maintainable and modular architecture

Architecture Overview

RetailPulse follows an end-to-end AI and MLOps workflow:

Raw Sales Data → Data Cleaning → Feature Engineering → Customer Segmentation & Churn Prediction → Forecasting Models (Prophet + LSTM + Hybrid Forecasting) → Inventory Optimization → Streamlit Dashboard → MLflow Tracking & Monitoring.



The system is modular, scalable, and follows MLOps practices such as experiment tracking, model monitoring, retraining readiness, and workflow automation.

Execution Timeline

Week 1 (25 June – 1 July 2026)

- Data collection and preprocessing
- Exploratory Data Analysis (EDA)
- Data cleaning and feature engineering
- Data visualization and initial insights

Week 2 (2 July – 8 July 2026)

- Advanced demand forecasting models
- Customer churn prediction using XGBoost
- Inventory optimization
- Hyperparameter tuning and model evaluation

Week 3 (9 July – 15 July 2026)

- Streamlit dashboard development
- Multi-page dashboard implementation
- Interactive analytics and KPI visualization
- Dashboard testing and refinement

Week 4 (16 July – 25 July 2026)

- GitHub Actions CI/CD pipeline
- Docker containerization
- Deployment and production testing
- Final documentation, project review, and report preparation

Technical Highlights

Modelling Techniques

- Sales trend analysis using Exploratory Data Analysis (EDA)
- Customer churn prediction using XGBoost
- Demand forecasting using Prophet and LSTM
- Hybrid forecasting model for improved prediction accuracy
- Inventory optimization using forecasting insights

Feature Engineering

- Data cleaning and missing value handling
- Customer segmentation and RFM feature creation
- Time-based feature extraction (day, month, year)
- Sales trend and seasonal feature generation
- Feature scaling and encoding for machine learning models

MLOps Practices

- MLflow for experiment tracking
- Optuna for hyperparameter tuning
- Data drift monitoring
- Airflow-based automated model retraining pipeline
- GitHub Actions for CI/CD automation
- Docker containerization for deployment

Dashboard & Deployment

- Multi-page Stream lit dashboard
- Interactive KPI and sales visualizations
- Demand forecasting dashboard
- Docker zed application deployment
- CI/CD pipeline integration using GitHub Actions

Challenges Faced

- Handling missing and inconsistent retail sales data
- Improving forecasting accuracy across different models
- Integrating multiple machine learning workflows
- Configuring deployment using Docker and GitHub Actions

Solutions

- Implemented a robust data preprocessing pipeline
- Used hybrid forecasting to improve prediction performance
- Built a modular and scalable project architecture
- Automated model training and deployment workflows
- Developed an interactive dashboard for real-time analytics

Deployment & Operation

Deployment Platform

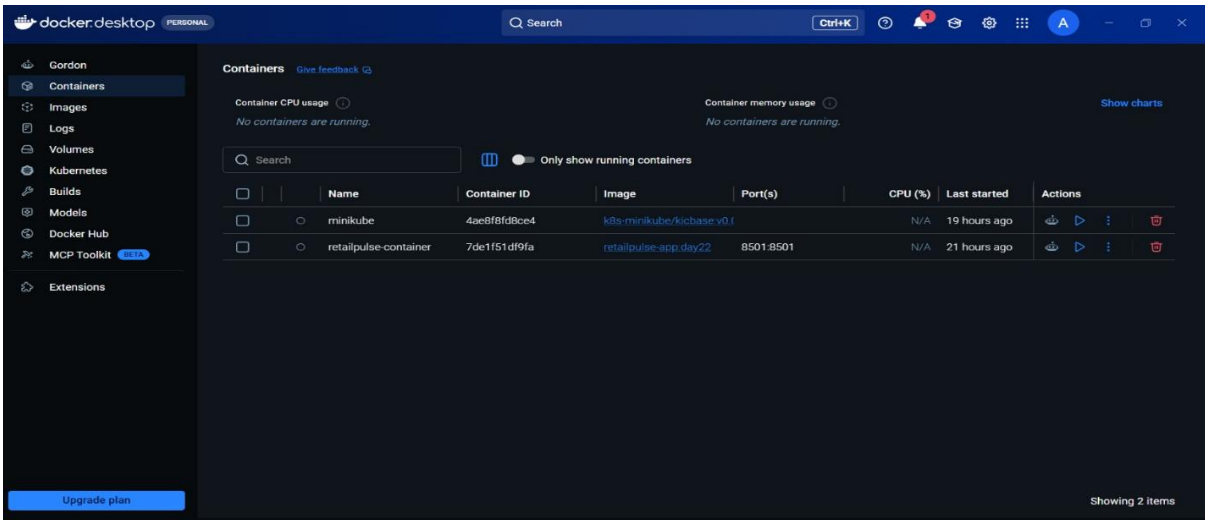
- Stream lit for interactive web application development
- Docker for containerized deployment
- GitHub for version control and collaboration
- GitHub Actions for CI/CD automation

CI/CD Summary

- Source code managed using GitHub repositories
- Automated build and workflow execution using GitHub Actions
- Docker-based containerization for consistent deployment
- Continuous integration and deployment support

Monitoring Setup

- MLflow for experiment tracking
- Data drift detection and monitoring
- Model performance evaluation
- Automated model retraining using Apache Airflow



Retail Pulse Dashboard stream lit :- https://retailpulse-56zyaydx7apms4d4jrkexh.streamlit.app/

Personal Reflection

This project provided valuable hands-on experience in applying Machine Learning and MLOps concepts to solve real-world retail analytics problems. Throughout the project, I gained practical knowledge in data preprocessing, exploratory data analysis, predictive modelling, dashboard development, and deployment. The project also strengthened my understanding of building scalable machine learning workflows and deploying applications using modern DevOps practices.

Key Learnings

- Hands-on experience with end-to-end Machine Learning workflows
- Data preprocessing, feature engineering, and model development
- Dashboard development using stream lit
- Practical implementation of demand forecasting, customer churn prediction, and inventory optimization
- CI/CD automation and containerized deployment using GitHub Actions and Docker

Industry Best Practices Applied

- Modular and reusable code architecture
- Version control and collaboration using GitHub
- Experiment tracking using MLflow
- Hyperparameter tuning using Optuna
- Docker zed deployment and CI/CD automation

Future Roadmap Ideas

- Real-time data integration and analytics
- Cloud-native deployment on AWS, Azure, or GCP
- Recommendation system integration
- Advanced automation, monitoring, and model retraining

Key Features

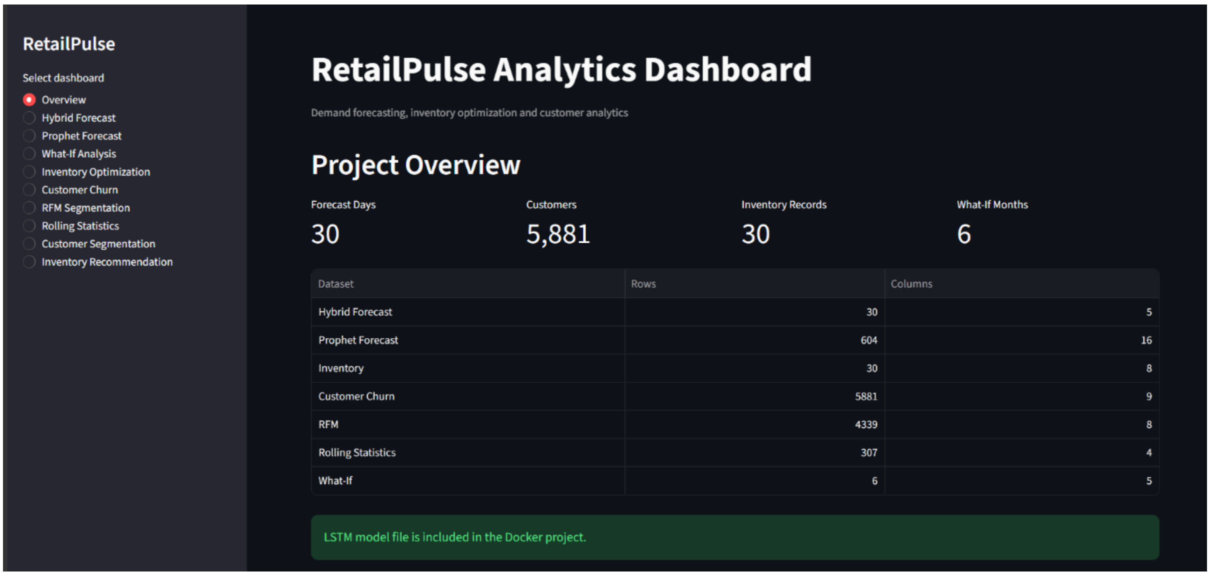
ID	Feature	Description	Acceptance Criteria
F1	Data Preprocessing & EDA	Cleans, preprocesses, and analyzes retail sales data	Clean dataset prepared with meaningful insights generated
F2	Customer Churn Prediction	Predicts customers likely to stop purchasing using XGBoost	Accurate churn predictions with evaluated model performance
F3	Demand Forecasting	Forecasts future product demand using Prophet and LSTM	Reliable demand forecasts generated for future periods
F4	Hybrid Forecasting	Combines Prophet and LSTM predictions to improve accuracy	Enhanced forecasting performance compared to individual models
F5	Inventory Optimization	Recommends inventory levels based on forecasted demand	Optimized stock planning with reduced overstock and stockout risk
F6	Streamlit Dashboard	Interactive dashboard for analytics, KPIs, and forecasts	Users can visualize and interact with business insights
F7	Data Drift Detection	Monitors changes in incoming data distribution	Drift alerts generated when significant data changes occur
F8	MLflow Experiment Tracking	Tracks model parameters, metrics, and experiment history	Experiment results logged and reproducible
F9	Airflow Pipeline Automation	Automates model retraining and workflow scheduling	Scheduled pipeline executes successfully
F10	CI/CD & Docker Deployment	Automates deployment using GitHub Actions and Docker	Application builds and deploys successfully with consistent environment

Technology Stack

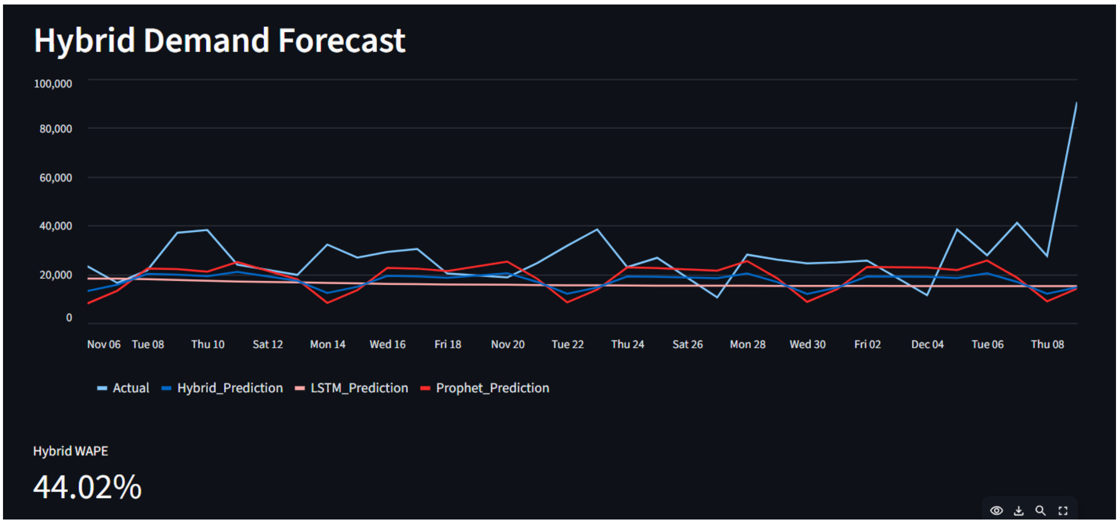
Category	Technology	Purpose
Programming	Python	Data analysis, machine learning, and automation
Frontend	Streamlit	Interactive dashboard development
Machine Learning	Scikit-learn, XGBoost	Customer churn prediction
Forecasting	Prophet, TensorFlow (LSTM)	Demand forecasting
MLOps	MLflow, Optuna, Apache Airflow	Experiment tracking, hyperparameter tuning, and workflow automation
Containerization	Docker	Portable application deployment
CI/CD	GitHub Actions	Automated build and deployment
Version Control	Git & GitHub	Source code management and collaboration

Visuals

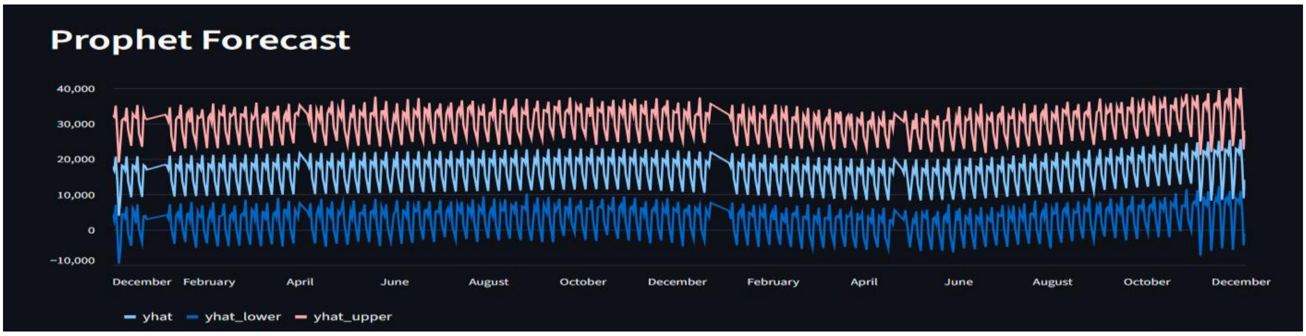
Dashboard Home Screen



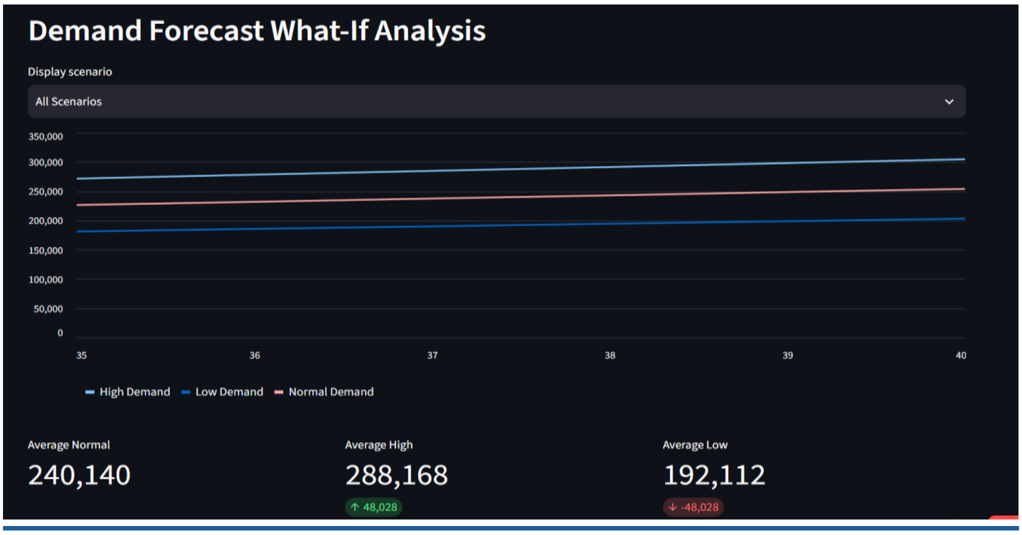
Hybrid Forecast



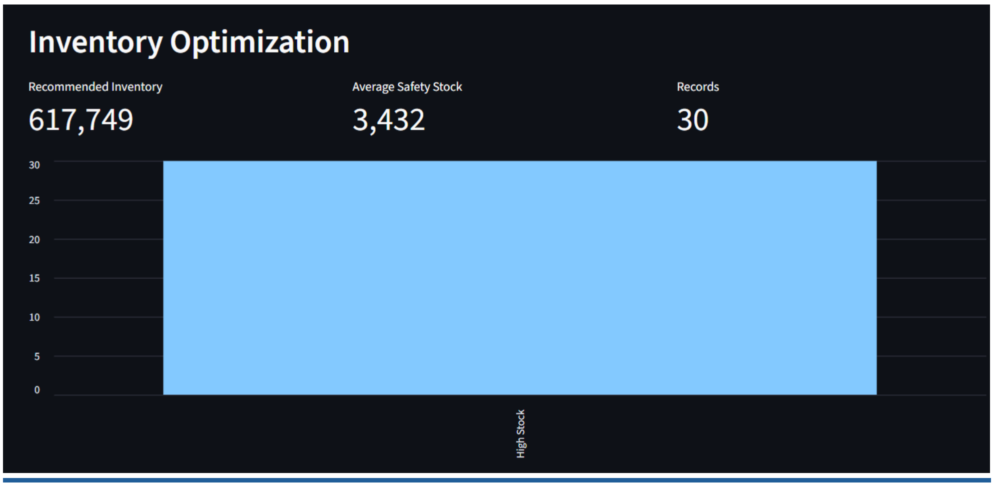
Prophet Forecast



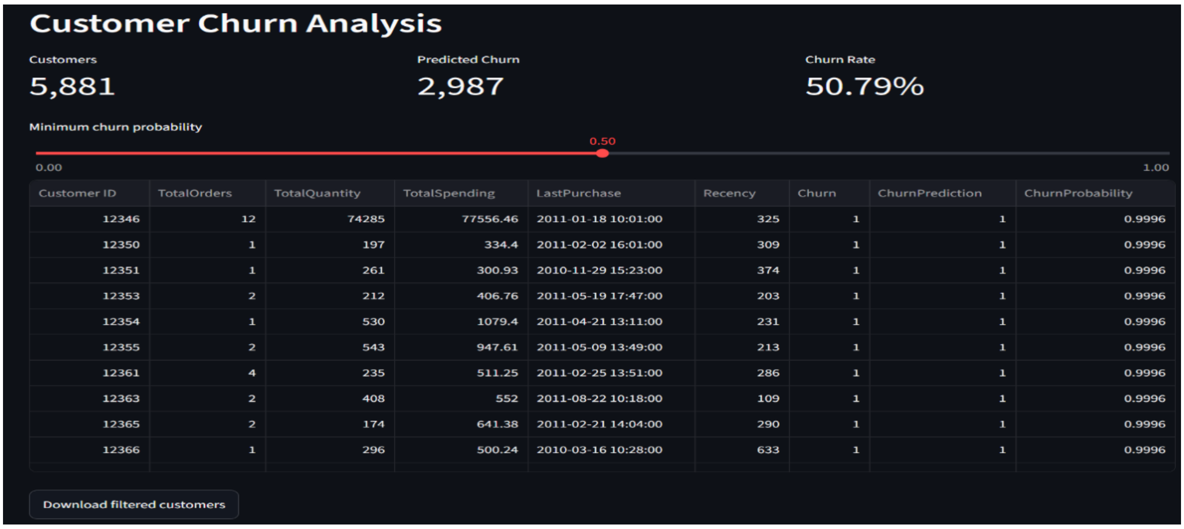
What if Analysis



Inventory Optimization

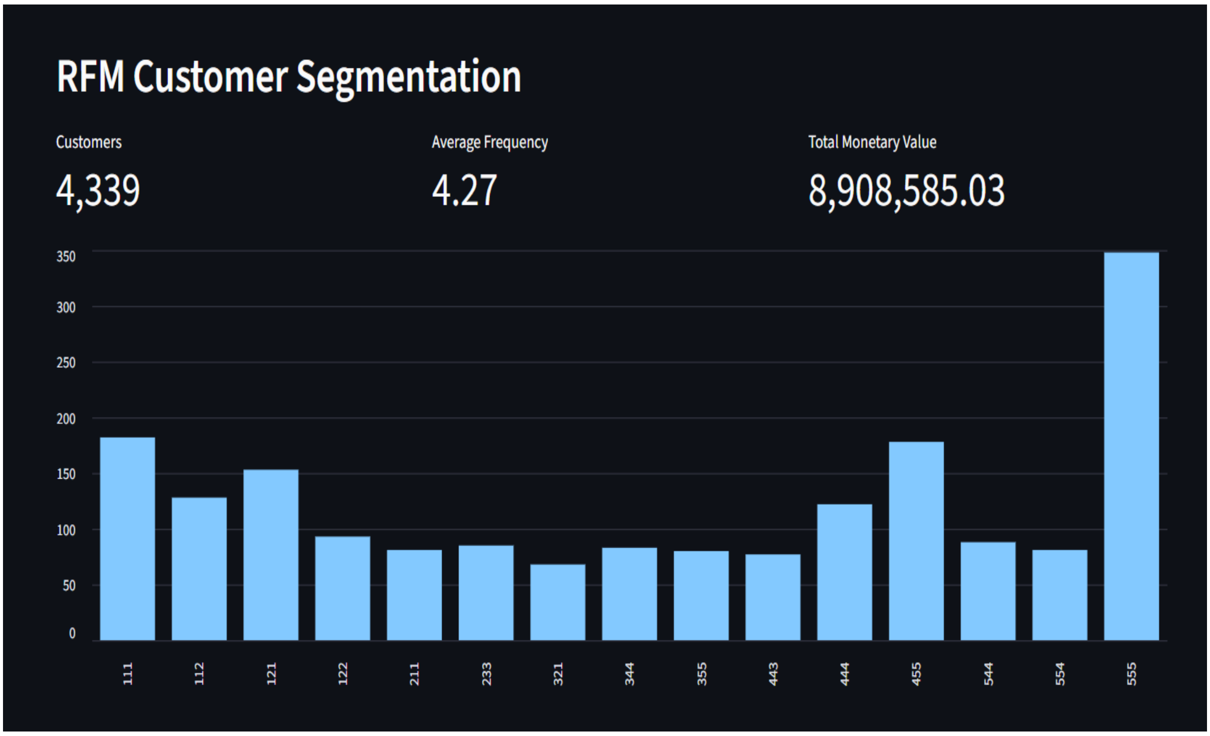


Customer Churn

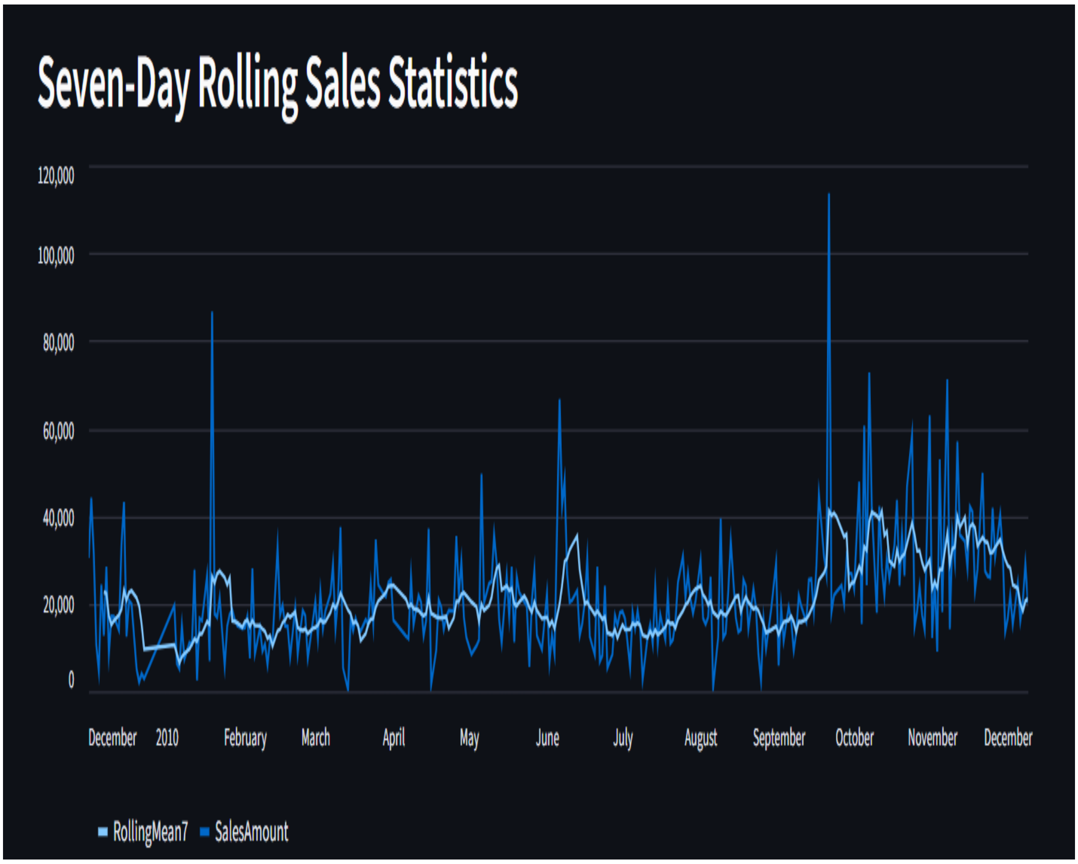


Customer ID	TotalOrders	TotalQuantity	TotalSpending	LastPurchase	Recency	Churn	ChurnPrediction	ChurnProbability
12346	12	74285	77556.46	2011-01-18 10:01:00	325	1	1	0.9996
12350	1	197	334.4	2011-02-02 16:01:00	309	1	1	0.9996
12351	1	261	300.93	2010-11-29 15:23:00	374	1	1	0.9996
12353	2	212	406.76	2011-05-19 17:47:00	203	1	1	0.9996
12354	1	530	1079.4	2011-04-21 13:11:00	231	1	1	0.9996
12355	2	543	947.61	2011-05-09 13:49:00	213	1	1	0.9996
12361	4	235	511.25	2011-02-25 13:51:00	286	1	1	0.9996
12363	2	408	552	2011-08-22 10:18:00	109	1	1	0.9996
12365	2	174	641.38	2011-02-21 14:04:00	290	1	1	0.9996
12366	1	296	500.24	2010-03-16 10:28:00	633	1	1	0.9996

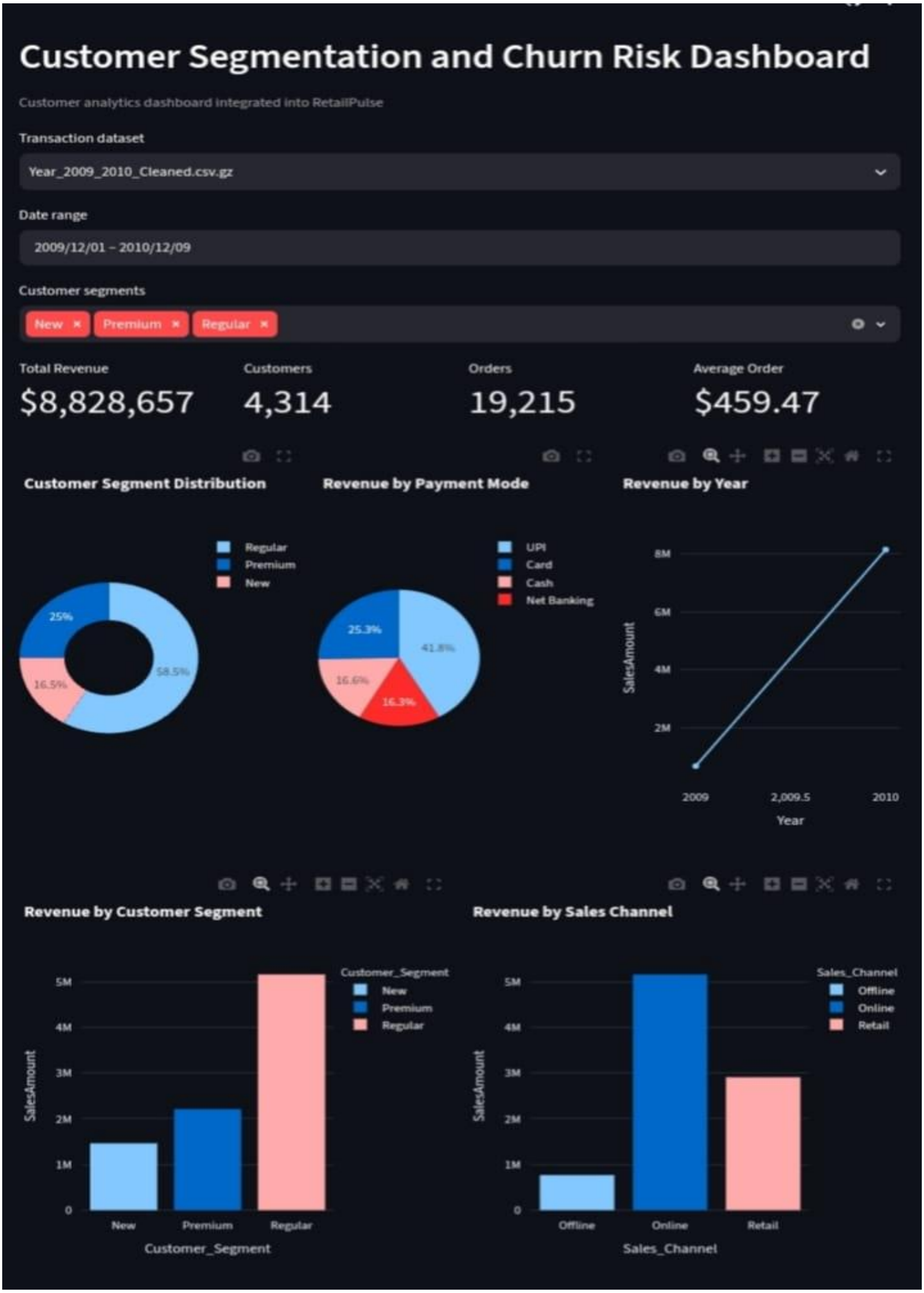
RFM Segmentation



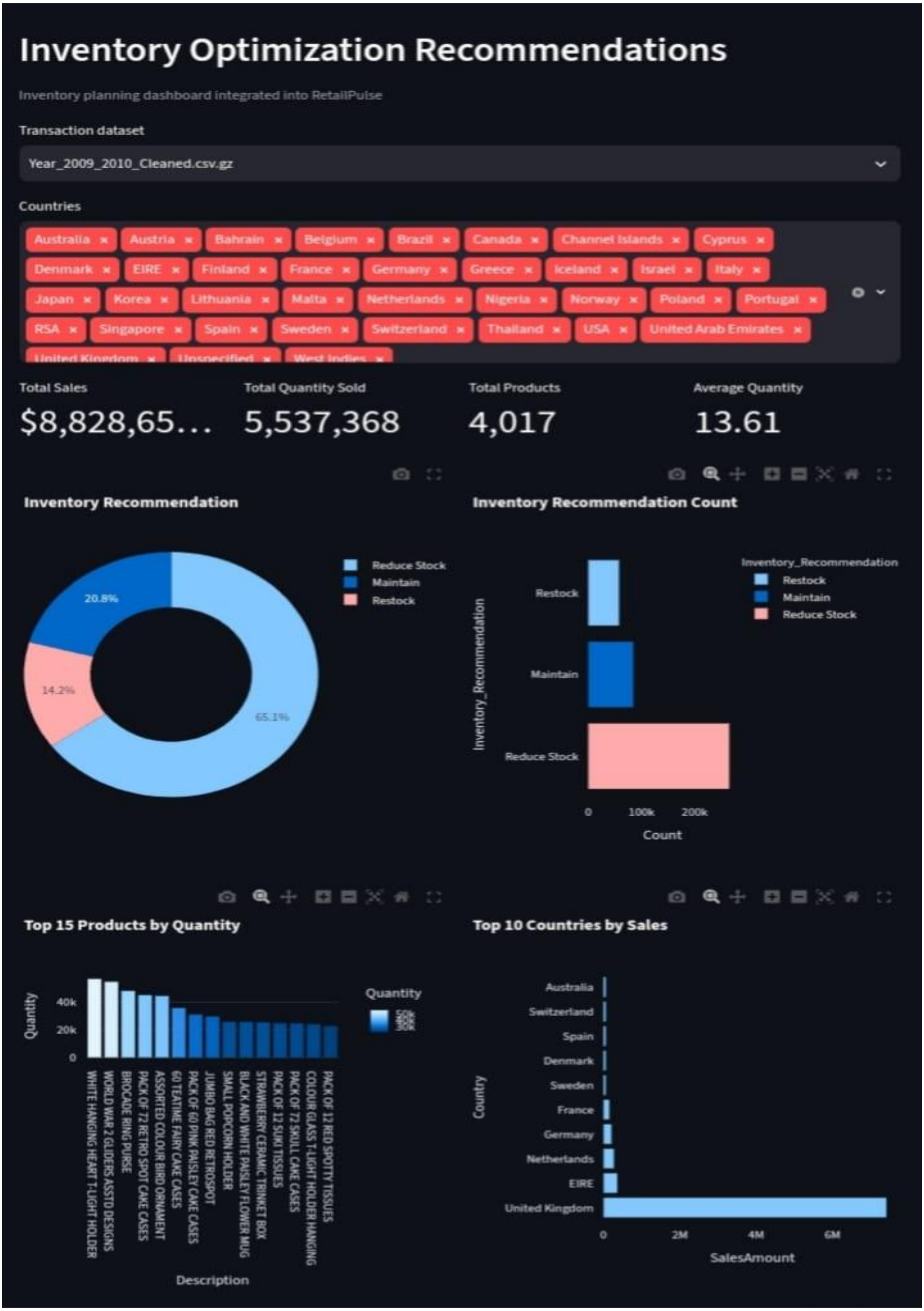
Rolling Statics



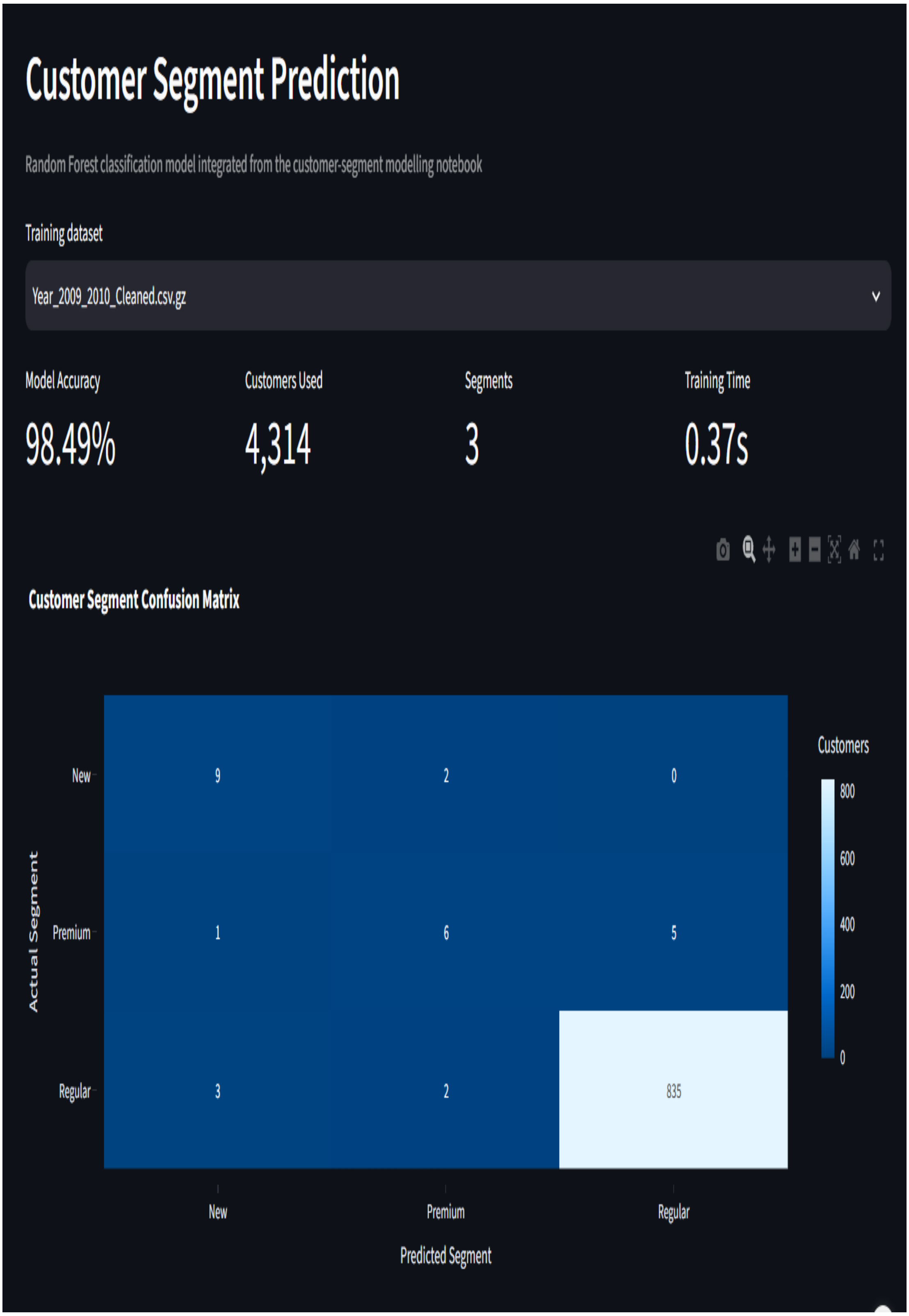
Customer Segmentation



Inventory Recommendation



Customer segmentation Prediction

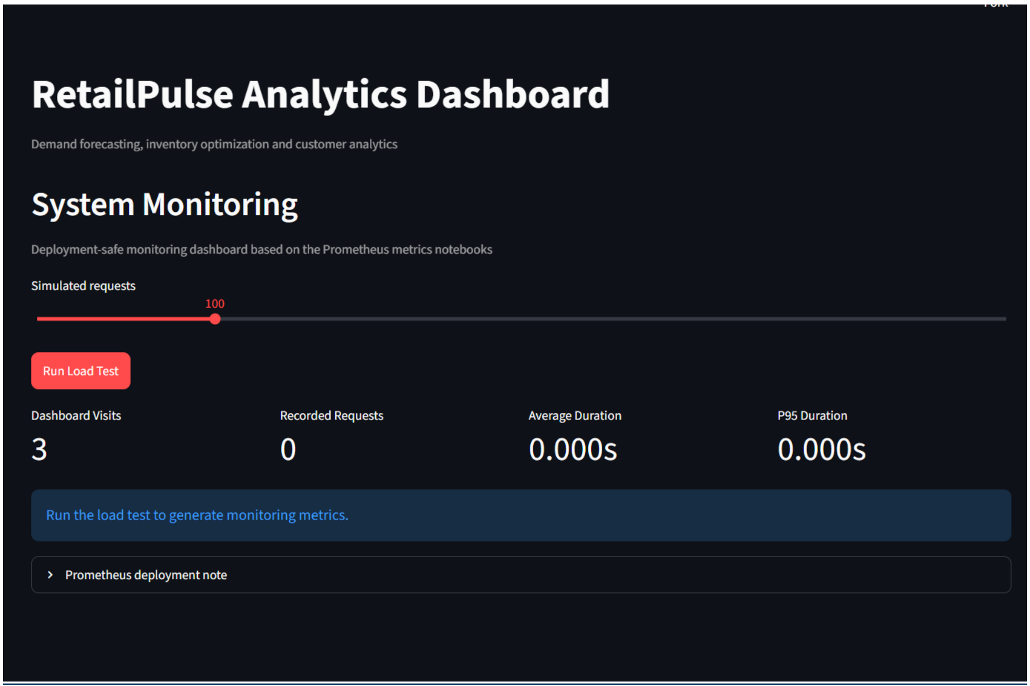


OUTPUT:

The Retail Demand Forecasting and Customer Churn Prediction System was successfully developed and deployed as an interactive stream lit dashboard. The application processes retail transaction data to generate meaningful business insights through machine learning models and visual analytics. Users can analyse historical sales trends, forecast future demand, identify customers at risk of churn, monitor inventory performance, and view key performance indicators (KPIs) in an intuitive interface. The dashboard provides dynamic charts, filtering options, and predictive results, enabling data-driven decision-making for retail businesses.

Key Outputs

- Interactive multi-page Streamlit dashboard.
- Demand forecasting with future sales predictions.
- Customer churn prediction using machine learning.
- Inventory analysis and optimization insights.
- Sales trend analysis with interactive visualizations.
- KPI dashboard for monitoring business performance.
- User-friendly interface for business analytics.



Conclusion

The Retail Demand Forecasting and Customer Churn Prediction project successfully demonstrates the application of data analytics and machine learning in solving real-world retail challenges. By integrating demand forecasting, customer churn prediction, inventory optimization, and interactive dashboards into a single platform, the project provides valuable insights that support informed business decisions. The use of Streamlit for visualization and modern machine learning techniques ensures that the system is both efficient and easy to use. Overall, the project highlights how predictive analytics can improve operational efficiency, reduce business risks, and enhance customer retention, making it a practical solution for modern retail management.